

TASK – 1 : DNA Sequence Analysis Using NCBI BLAST

Objective

The aim of this task was to retrieve the human Hemoglobin Beta (HBB) gene sequence from the NCBI database and perform BLAST analysis to find homologous sequences and study sequence similarity.

Introduction

Bioinformatics tools are widely used for the analysis of biological sequences. BLAST (Basic Local Alignment Search Tool) is one of the most important tools used to compare DNA or protein sequences against biological databases. In this experiment, the human Hemoglobin Beta (HBB) gene sequence was obtained from NCBI and analyzed using BLAST to determine sequence similarity and alignment statistics.

Methodology

- The NCBI Gene database was opened.
- The gene "HBB (Hemoglobin Beta) Homo sapiens" was searched.
- The nucleotide sequence with accession number NM_000518.5 was selected.
- The sequence was retrieved in FASTA format.
- BLAST analysis was performed using the NCBI BLAST tool.
- Alignment results and sequence statistics were recorded.

Results

The screenshot displays the NCBI Gene database search results for the HBB (Hemoglobin Beta) gene in Homo sapiens. The search bar at the top shows "HBB Homo sapiens" with a "Search" button. Below the search bar, the results are displayed in a tabular format, sorted by Relevance. The main entry is "HBB -- hemoglobin subunit beta" from Homo sapiens (human), with Gene ID 3043. The entry is also known as CD113t.C, ECT6, beta-globin. The results are categorized by "RefSeq products", "Orthologs", and "Genome Data Viewer". A "New - Visualize gene across multiple species" link is visible. The "RefSeq Sequences" section is expanded, showing the sequence details. The "Filters" section on the right shows "Results by taxon" with a list of top organisms, including Homo sapiens (219), Severe acute respiratory syndrome coronavirus 2 (3), Sporothrix schenckii 1099-18 (2), Scodosporium apiospermum (1), and Fonsecaea monophora (1). The "Find related data" section shows a dropdown menu for "Database" and a "Find items" button. The "Search details" section shows the search criteria: "(HBB[All Fields] AND ('Homo sapiens'[Organism] OR Homo sapiens[All Fields])) AND alive[prop]".

Figure 1 : Search results for HBB(Hemoglobin Beta) gene in NCBI Gene database

Download

 datasets

API

Gene name	hemoglobin subunit beta
Gene symbol	HBB
GeneID	3043
Organism	Homo sapiens (human)
Gene type	protein-coding
Synonyms	ECYT6, CD113t-C, beta-globin
Proteins	1
Transcripts	1
MANE Select	NM_000518.5
RefSeq summary	The alpha (HBA) and beta (HBB) loci determine the structure of the 2 types of polypeptide chains in adult hemoglobin, Hb A. The normal adult hemoglobin tetramer consists of two alpha chains and two beta chains. Mutant beta globin causes sickle cell anemia. Absence of beta chain causes beta-zero-thalassemia. Reduced amounts of detectable beta globin causes beta-plus-thalassemia. The order of the genes in the beta-globin cluster is 5'-epsilon -- gamma-G -- gamma-A -- delta -- beta--3'. [provided by RefSeq, Jul 2008]

View less ^

Figure 2: Gene information page of the HBB (Hemoglobin Beta) gene obtained from NCBI Gene database.

National Library of Medicine
National Center for Biotechnology Information

Log in

Nucleotide

Advanced

Help

FASTA

Send to:

Change region shown

Customize view

Analyze this sequence
Run BLAST
Pick Primers
Show in Genome Data Viewer

More about the HBB gene
The alpha (HBA) and beta (HBB) loci determine the structure of the 2 types of polypeptide chains in adult hemoglobin, Hb A. The normal adult ...
Also Known As: CD113L-C, ECYT6, beta-g...

Related information
Protein
PubMed
Taxonomy
Annotated Genomic

Homo sapiens hemoglobin subunit beta (HBB), mRNA
NCBI Reference Sequence: NM_000518.5
[GenBank](#) [Graphics](#)

```

>NM_000518.5 Homo sapiens hemoglobin subunit beta (HBB), mRNA
ACATTTGCTTCTGACCAACTGTGTTCACTAGCAACCTCAAAACAGACCATGGTGCATCTGACTCCTGA
GGAGAAGCTGCGCTTACTGCCCTGTGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGC
AGGCTCGTGGTGGTCTACCTTGGACCCAGAGGTTCTTTGAGTCCCTTTGGGATCTGTCCACTCCTGATG
CTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCCTTAGTGATGGCCTGGC
TCACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCATGTGACAAGCTGCACGTGGAT
CCTGAGAAGTTCAAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAGAATTCA
CCCCACAGTGCAGGCTGCTATCAGAAAGTGGTGGTGGTGGTAAATGCCCTGGCCACAAGTATCA
CTAAGCTCGCTTTCTTGTGCTCCAAATTTCTATTAAGGTTCTTTGTTCCCTAAGTCCAACACTAAACT
GGGGGATATTATGAAGGGCTTGAAGCATCTGGATTCTGCCTAATAAAAAACATTATTTCATTGCAA

```

Figure 3: FASTA format of the HBB nucleotide sequence retrieved from the NCBI database.

BLAST® » blastn suite Home Recent Results Saved Strategies Help

blastn blastp blastx tblastn tblastx **Standard Nucleotide BLAST**

BLASTn programs search nucleotide databases using a nucleotide query. more...

Enter Query Sequence Reset page Bookmark

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#) Query subrange [?](#)

NM_000518.5 From

To

Or, upload file No file chosen [?](#)

Job Title

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database ☒ Standard databases (nr etc.): ☐ rRNA/ITS databases ☐ Genomic + transcript databases ☐ Betacoronavirus ☐ Experimental databases

Organism Optional

Enter organism name or id—completions will be suggested ☐ exclude

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Exclude Optional

☐ Models (XM/XP) ☐ Uncultured/environmental sample sequences

Limit to Optional

☐ Sequences from type material

Entrez® Query Optional

Enter an Entrez query to limit search [?](#) [YouTube](#) Create custom database

Program Selection

Optimize for ☒ Highly similar sequences (megablast) ☐ More dissimilar sequences (discontinuous megablast)

Figure 4: BLAST input page showing the HBB sequence used for sequence similarity analysis.

Descriptions	Graphic Summary	Alignments	Taxonomy					
Sequences producing significant alignments								
Download Select columns Show 100								
select all 100 sequences selected								
GenBank Graphics Distance tree of results MSA Viewer								
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Homo sapiens hemoglobin subunit beta (HBB) mRNA	Homo sapiens	1160	1160	100%	0.0	100.00%	628	NM_000518.5
PREDICTED: Pan troglodytes hemoglobin subunit beta (HBB) mRNA	Taxonomy for Homo sapiens	1155	1155	100%	0.0	99.84%	639	XM_508242.5
PREDICTED: Pan paniscus hemoglobin subunit beta (LOC100976465) mRNA	Pan paniscus	1149	1149	100%	0.0	99.68%	643	XM_003819029.5
Homo sapiens hemoglobin beta mRNA (cDNA clone MGC:14540 IMAGE 429212 ...)	Homo sapiens	1142	1142	100%	0.0	99.52%	658	BC007075.1
Human messenger RNA for beta-globin	Homo sapiens	1140	1140	100%	0.0	99.52%	626	V00497.1
PREDICTED: Gorilla gorilla gorilla hemoglobin subunit beta (LOC101126932) mRNA	Gorilla gorilla	1133	1133	100%	0.0	99.20%	638	XM_019036164.3
Homo sapiens hemoglobin beta mRNA complete cds	Homo sapiens	1133	1133	100%	0.0	99.36%	647	AY509193.1
PREDICTED: Pongo pygmaeus hemoglobin subunit beta (HBB) mRNA	Pongo pyg...	1105	1105	100%	0.0	98.41%	639	XM_054440830.2
PREDICTED: Pongo abelii hemoglobin subunit beta (HBB) mRNA	Pongo abelii	1105	1105	100%	0.0	98.41%	658	XM_002822127.6
PREDICTED: Symphalangus syndactylus hemoglobin subunit beta (HBB) mRNA	Symphalang...	1099	1099	100%	0.0	98.25%	638	XM_055282707.1
PREDICTED: Nomascus leucogenys hemoglobin subunit beta (HBB) mRNA	Nomascus l...	1088	1088	100%	0.0	97.93%	753	XM_004090649.3
PREDICTED: Hylobates moloch hemoglobin subunit beta (HBB) mRNA	Hylobates ...	1077	1077	100%	0.0	97.61%	638	XM_032166808.2
Homo sapiens hemoglobin beta chain variant (HBB) mRNA_HBB-Dothan allele, co...	Homo sapiens	1064	1064	94%	0.0	99.32%	587	EU694432.1
Homo sapiens hemoglobin beta subunit variant (HBB) mRNA complete cds	Homo sapiens	1064	1064	92%	0.0	99.83%	579	AF181989.1
Homo sapiens beta globin chain variant (HBB) mRNA complete cds	Homo sapiens	1057	1057	93%	0.0	99.48%	581	AF349114.1
PREDICTED: Trachypithecus francoisi hemoglobin subunit beta (LOC117081249) ...	Trachypithe...	1038	1038	100%	0.0	96.51%	749	XM_033207068.1

Figure 5: BLAST results table displaying homologous sequences and alignment statistics obtained for the HBB sequence.

Homo sapiens hemoglobin subunit beta (HBB), mRNA

Sequence ID: [NM_000518.5](#) Length: 628 Number of Matches: 1

Range 1: 1 to 628 [GenBank](#) [Graphics](#)

▼ [Next Match](#) ▲ [Previous Match](#)

Score	Expect	Identities	Gaps	Strand
1160 bits(628)	0.0	628/628(100%)	0/628(0%)	Plus/Plus
Query 1	ACATTTGCTTCTGACACAACCTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATC			
Sbjct 1	ACATTTGCTTCTGACACAACCTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATC			
Query 61	TGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAG			
Sbjct 61	TGACTCCTGAGGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAG			
Query 121	TTGGTGGTGAGGCCCTGGGCAGGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTG			
Sbjct 121	TTGGTGGTGAGGCCCTGGGCAGGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTG			
Query 181	AGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAACCCTAAGGTGAAGGCTC			
Sbjct 181	AGTCCTTTGGGGATCTGTCCACTCCTGATGCTGTTATGGGCAACCCTAAGGTGAAGGCTC			
Query 241	ATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCTCAAGG			
Sbjct 241	ATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGCTCACCTGGACAACCTCAAGG			
Query 301	GCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACT			
Sbjct 301	GCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGATCCTGAGAACT			
Query 361	TCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCA			
Sbjct 361	TCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCA			
Query 421	CCCCACCAGTGCAGGCTGCCTATCAGAAAAGTGGTGGCTGGTGTGGCTAATGCCCTGGCCC			

Figure 6: Pairwise alignment of the HBB sequence showing 100% sequence identity, alignment score, and E-value obtained from BLAST analysis.

Sequence Information

- Gene Name: Hemoglobin Subunit Beta (HBB)
- Organism: Homo sapiens
- Accession Number: NM_000518.5
- Sequence Length: 628 bp

BLAST Alignment Results

Parameter	Value
Score	1160 bits (628)
E-value	0.0
Identity	628/628 (100%)
Gaps	0/628 (0%)
Strand	Plus/Plus

Observation

The BLAST analysis produced a perfect alignment between the query sequence and the subject sequence. The alignment score was 1160 bits with an E-value of 0.0, indicating a highly significant match. The sequence identity was 100%, and no gaps were observed in the alignment.

Discussion

BLAST results demonstrated complete sequence similarity between the query sequence and the matched database sequence. The absence of gaps and the 100% identity indicate that the sequence is highly conserved and accurately represented in the NCBI database.

Conclusion

The HBB gene encodes the beta chain of hemoglobin, a protein responsible for oxygen transport in red blood cells. The

The human Hemoglobin Beta (HBB) gene sequence was successfully retrieved from the NCBI database and analyzed using BLAST. The sequence showed a perfect alignment with 100% identity, no gaps, and an E-value of 0.0. The study demonstrates the usefulness of BLAST in identifying homologous sequences and performing sequence comparison in bioinformatics research.